

Computational Interaction Science

Foundations of a Scientific Theory of Computational Interaction

Final synthesis of the foundational research program

Final determination: Computational Interaction Science is best understood as an interdisciplinary synthesis.

Abstract

This paper integrates a sequence of foundational studies concerning scientific objects, representation, ontology, semantics, interpretation, opportunity, interaction, behavior, measurement, similarity, transfer, strategy, formal modeling, prediction, and comparative evaluation. Its purpose is to determine whether the accumulated evidence justifies Computational Interaction Science as an independent scientific discipline. The synthesis finds that computational interaction is a legitimate primary explanandum with a coherent ontology, explicit boundary conditions, formal representations, measurement contracts, falsifiable hypotheses, and a cumulative empirical program. The framework also contributes real explanatory organization by separating target from representation, semantic structure from interpretation, system-relative opportunity from agent-relative belief, interaction from behavior, records from evidence, and similarity from equivalence. However, the principal mechanisms remain substantially inherited from established disciplines: coupling and feedback from cybernetics and control; perception-action relations from ecological psychology; cognition and learning from cognitive science; formal transition and process semantics from computer science; authorization from security; measurement from psychometrics and metrology; and situated, distributed, and institutional organization from HCI, activity theory, and social science. The program has not yet demonstrated stable laws, unique mechanisms, autonomous methods, or predictive gains that cannot be obtained by disciplined integration of these fields. The evidence therefore supports Computational Interaction Science as an interdisciplinary scientific synthesis and research program, not yet as an independent discipline.

1. Problem and Decision Rule

The central question is not whether computational interaction is important, measurable, or theoretically rich. It is whether scientific understanding materially improves by recognizing a distinct discipline rather than treating the subject as an integrated research program within and across existing sciences.

The decision must separate three claims that are often conflated: interaction can be a scientific object; a coherent theory of interaction can exist; and an autonomous discipline is necessary. The first two may be true while the third remains unsupported.

The discipline claim is accepted only if the research program exhibits a sufficiently distinct explanatory object, ontology, mechanisms, formal and empirical methods, predictive or explanatory gains, cumulative problem structure, and boundaries that cannot be supplied adequately through integration of existing disciplines. Mere terminological unity, institutional convenience, or breadth of application is insufficient.

2. Method of Synthesis

The synthesis treats each preceding paper as a constrained contribution rather than as independent proof of the final conclusion. Definitions, formal contracts, and empirical claims were compared for consistency; duplicated constructs were normalized to established vocabulary; conflicts were retained where they mark genuine theoretical alternatives; and each proposed novelty was tested against reduction to adjacent theories.

The final falsification test asks whether the complete framework can be reconstructed from existing disciplines without significant explanatory loss. If it can, a new discipline is not scientifically necessary. If integration fails because a distinctive object requires unique mechanisms, methods, or predictions, disciplinary independence becomes defensible.

- Object test: Is computational interaction a stable and discriminating explanatory target?
- Ontology test: Does the framework define a coherent set of entities and relations without unnecessary duplication?
- Mechanism test: Are any generating mechanisms distinctive rather than borrowed or recombined?
- Measurement test: Can constructs be operationalized with reproducible evidence and uncertainty?
- Formal test: Can the theory be represented compositionally and analyzed?
- Prediction test: Does it support scoped, testable forecasts and intervention claims?
- Necessity test: Does the synthesis explain phenomena that strong integrated baselines cannot?

3. Explanatory Object

The primary explanatory object is not the interface, the user, the device, the software product, or isolated behavior. It is the computational interaction episode: a bounded interval in which distinguishable systems are coupled such that

perturbations alter another system's possible or actual trajectory and the resulting configuration reciprocally or recurrently constrains a later trajectory within a declared boundary and timescale.

This definition is strict enough to distinguish interaction from one-way causation, isolated action, passive observation, mere co-presence, and arbitrary input-output succession. Reciprocity may be asymmetric, delayed, probabilistic, environmentally mediated, or distributed over a sequence. Intentionality and communication are not necessary.

Behavior is downstream but not synonymous: it is a bounded, temporally extended trajectory of observable or instrumentally recoverable changes, outputs, or regulated persistence attributed to a system under a declared observation model. A record is not behavior until detection, attribution, boundary selection, and interpretation have been made explicit.

4. Complete Ontology

Concept	Canonical role
Domain	Declared entities, relations, constraints, processes, and possible states relevant to the study.
Participant	A distinguishable system assigned a role within an interaction boundary.
Environment	Relevant conditions and structures not assigned to a participant but participating in coupling.
State	A time-indexed assignment of relevant properties under a declared model.
Action	A bounded intervention or attributed act capable of affecting a trajectory.
Coupling	A dependency through which one system constrains another system's possible or actual trajectory.
Interaction episode	A bounded interval of reciprocal or recurrent constraint through coupling.
Semantic structure	Denotational, inferential, truth-conditional, or transition-relevant relations assigned to states or expressions.
Representation	A realized vehicle providing selective access to a target through mapping and interpretive practice.
Projection	An experimentally or analytically declared transformation from target structure to an observer-accessible form.
Observation	An instrument- and model-relative registration of selected aspects of an episode.
Interpretation	Contextual application or reconstruction of semantic relations by an agent or analytical process.
Opportunity profile	A coordinated typed description of possible, capable, permitted, enabled, reachable, exposed, signaled, believed, and recoverable action relations.
Behavioral trace	A temporally or causally ordered record of attributed events under a declared observation model.
Strategy	An inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response.
Experience state	Persistent changes in knowledge, skill, habit, calibration, artifact use, or coordination attributable to prior episodes.
Transfer	A source-dependent effect expressed in a distinct target context relative to a suitable counterfactual baseline.
Measurement	A rule-governed mapping from observations to values, estimates, or construct claims with declared uncertainty and admissible interpretation.
Similarity profile	A typed set of identity, equivalence, distance, or incomparability claims under declared observables and transformations.
Outcome	A state, criterion value, or consequence used to evaluate an episode or strategy.
Evidence	A traceable relation among records, observations, coding, estimates, assumptions, alternatives, and claims.

5. Dependency Architecture

The framework is relational rather than a one-way pipeline. Nevertheless, an analytical dependency order is required to prevent category errors:

Domain and ontology -> semantic structure -> representation and projection
-> observation and interpretation -> opportunity profile -> interaction episode
-> behavioral trace -> strategy and adaptation -> measurement, comparison, and prediction

Feedback operates in the reverse direction. Behavior changes state; outcomes change beliefs and strategies; repeated interaction changes experience; institutions alter permissions and representations; measurement choices alter what becomes visible as evidence. The arrows therefore mark analytical dependence, not deterministic causation.

6. Foundational Assumptions

1. Targets and representations are distinguishable. A representation is never assumed identical to what it represents.
2. Equivalence is typed. Semantic, informational, structural, opportunity, behavioral, strategic, and measurement equivalence are distinct relations.
3. Interaction boundaries, participants, state variables, and timescales are analytical commitments and must be declared.
4. Representation does not determine behavior. Effects are mediated by interpretation, opportunity, goals, capabilities, history, and context.
5. System-relative availability and agent-relative belief are different. Feasibility, permission, exposure, signaling, belief, execution, and success must be separated.
6. Records are not self-interpreting. Observation, event detection, attribution, segmentation, and construct inference require explicit models.
7. Latent constructs are hypotheses supported by converging evidence, not properties read directly from traces.
8. Prediction is scoped and probabilistic. Predictability depends on target, horizon, information set, population, context, model class, and tolerated error.
9. Comparison requires a declared common frame. Incomparability is a valid result.
10. Existing terminology and mechanisms are preferred unless explanatory loss can be demonstrated.

7. First Principles

Principle	Statement
Coupled constraint	Interaction exists when distinguishable systems reciprocally or recurrently constrain possible or actual trajectories through coupling within a declared episode.
Selective access	Every projection preserves, transforms, omits, emphasizes, or introduces access to target properties.
Typed opportunity	Action availability is not one set; it is a family of relations concerning feasibility, capability, authority, state, access, signal, belief, consequence, and recovery.
Situated interpretation	Meaningful uptake depends on context, prior experience, goals, conventions, and available evidence.
Trace discipline	Scientific claims must preserve the chain from records to observations, attributed events, measures, constructs, and explanations.
Compositionality with boundaries	Episodes may compose sequentially, concurrently, hierarchically, or through shared state only when boundary and interface conditions are explicit.
Typed comparison	Identity, equivalence, similarity, and ranking require declared observables, transformations, metrics, tolerances, and uncertainty.
Experience dependence	Prior episodes can change future beliefs, skills, habits, strategies, and coordination, but transfer must be established against a counterfactual baseline.
Bounded prediction	Near-term and tightly constrained phenomena are generally more predictable than latent, long-horizon, adaptive organization.
Comparative explanatory gain	The theory earns scientific necessity only if it outperforms strong integrated accounts in explanation, prediction, intervention, or unification.

8. Generative and Explanatory Mechanisms

The framework does not posit a single interaction mechanism. It organizes a family of established mechanisms operating at different levels.

Mechanism family	Examples
Dynamical	Coupling, feedback, control, synchronization, perturbation, stability, and recovery.
Computational	State transition, protocol execution, scheduling, concurrency, resource contention, and authorization.
Ecological	Perception of structured information, effectivities, affordances, and calibration.

Mechanism family	Examples
Cognitive	Attention, memory, inference, learning, decision, planning, confidence, and strategy selection.
Representational	Encoding, abstraction, salience, externalization, transformation, and distributed computation.
Social and institutional	Norms, roles, authority, language, convention, division of labor, and organizational memory.
Developmental	Skill acquisition, habit formation, transfer, adaptation, and strategy evolution.
Measurement	Instrumentation, event detection, coding, estimation, aggregation, and uncertainty propagation.

This breadth gives the synthesis coverage but weakens the case for disciplinary independence: the mechanisms are not uniquely interaction-scientific. The distinctive contribution is their disciplined coordination around a shared explanandum.

9. Formal Architecture

The recommended formal foundation is a layered observation-enriched probabilistic labeled transition system rather than a new calculus.

$$M = \langle N, S, A, T, I, F, C, P, X, \text{Sigma}, O, Y, B, G, \text{Mu}, R \rangle$$

- N: participants and roles; S: factored system and environment states; A: typed actions and events; T: deterministic or probabilistic transition relation.
- I and F: episode initiation and completion conditions; C: capability; P: permission; X: projection exposure; Sigma: signaling.
- O: participant observation model; Y: instrument record model; B: belief state; G: goals or evaluation criteria; Mu: strategy or policy; R: recovery and reversibility relations.
- Process-algebra operators supply composition, synchronization, hiding, and relabeling; event structures or Petri-net views preserve concurrency, conflict, and resource structure.
- Equivalence and similarity remain typed: trace, simulation, bisimulation, testing, semantic, opportunity, behavioral, strategic, outcome, and measurement-relative relations are not interchangeable.

The architecture is sufficient as a research representation because it separates system dynamics, projection, epistemic state, action selection, measurement, and recovery while permitting mappings to established formalisms. It is not evidence of a novel mathematical foundation.

10. Evidence and Measurement Framework

The empirical system of record is a three-contract chain:

Interaction Contract -> Behavioral Evidence Contract -> Measurement Contract

- The Interaction Contract declares participants, boundaries, states, coupling, preconditions, events, transitions, protocol, invariants, failure, recovery, completion, and uncertainty.
- The Behavioral Evidence Contract declares records, detection rules, attribution, segmentation, trace ordering, alternative explanations, and inferential limits.
- The Measurement Contract declares construct, measurand, unit, instrument, operationalization, transformation, scale, uncertainty, validity evidence, population, context, and prohibited interpretation.

The canonical evidence ladder is: raw record; verified observation; coded observation; derived measure; model-based estimate; construct evidence; mechanism evidence; generalized claim. Every upward transition requires additional assumptions and possible falsifiers.

11. Prediction

Computational interaction supports predictive science only in a bounded sense. Low-level and well-specified phenomena such as motor time, choice under constrained alternatives, practiced task time, near-term transitions, and localized risks can support quantitative forecasts. Discovery, error, recovery, and abandonment are predictable probabilistically when state, task, projection, history, and population are adequately represented.

Long-horizon adaptation, transfer, expertise, and strategy are weaker targets because latent state, nonstationarity, multiple viable policies, social context, and changing opportunity structures compound uncertainty. Predictive success does not establish mechanism, and intervention claims require causal assumptions beyond observational accuracy.

A valid prediction must declare target, horizon, information set, population, context, model class, baseline, uncertainty, calibration, transport scope, and decision consequence.

12. Methodological Framework

11. Declare the target phenomenon, episode boundary, participants, timescale, and explanatory question.
12. Specify ontology, semantic commitments, state variables, and admissible transitions.
13. Describe each representation or projection using a projection contract.
14. Construct typed opportunity profiles for relevant participants and states.
15. Collect records under an explicit observation and instrumentation model.
16. Detect and attribute behavioral events without importing latent explanations prematurely.
17. Operationalize constructs using measurement contracts and uncertainty budgets.
18. Compare interactions only after passing target, boundary, correspondence, measurement, and population comparability gates.
19. Fit explanatory and predictive models against strong domain-specific and integrated baselines.
20. Test transfer, strategy, and causal claims with appropriate counterfactual or intervention designs.
21. Report typed profiles, uncertainty, failures, and incomparability rather than forcing universal scores.
22. Update the shared ontology only when an irreducible explanatory gap has been demonstrated.

13. Empirical Program

Program	Core test
Projection effects	Hold semantic target and feasible operations fixed; manipulate exposure or signaling; measure discovery, belief calibration, selection, error, and recovery.
Opportunity model gain	Compare models with and without typed opportunity variables on held-out behavior and recovery prediction.
Interaction boundary reliability	Test whether independent analysts identify comparable episode boundaries and coupling closure from the same records.
Trace attribution validity	Compare event and action coding across instruments, coders, and alternative observation models.
Similarity validity	Test whether typed similarity profiles predict transfer and performance better than implementation labels or aggregate distance.
Transfer mechanisms	Manipulate source-target structural relations; measure carrier persistence, accessibility, positive transfer, negative transfer, and far transfer.
Strategy inference	Validate inferred strategies against perturbations, counterfactual tasks, verbal evidence, and out-of-sample traces.
Formal-model adequacy	Evaluate whether the layered transition architecture represents observed concurrency, uncertainty, authorization, projection, and recovery without ad hoc exceptions.
Predictive boundaries	Map forecast accuracy and calibration across horizons, populations, novelty, and nonstationarity.
Comparative decision value	Test whether gate-and-profile evaluation produces more reliable design or policy decisions than aggregate usability scores.

14. Relationship to Adjacent Sciences

Adjacent science	Relationship
Human-Computer Interaction	Primary institutional home; contributes interaction concepts, methods, usability, design, evaluation, and socio-technical analysis. The synthesis is narrower in explanatory ambition and broader in implementation independence.
Human Factors	Supplies performance, workload, error, safety, ergonomics, and systems evaluation. The synthesis adds explicit representation, opportunity, and interaction contracts.
Cognitive Science and Psychology	Supply perception, memory, reasoning, learning, decision, expertise, and skill mechanisms. The synthesis does not replace these mechanisms.
Ecological Psychology	Supplies agent-environment relations, affordances, and ecological information. It does not by itself encode authorization, projection, formal

Adjacent science	Relationship
	state, or measurement layers.
Cybernetics and Control	Supply coupling, feedback, regulation, stability, and controllability. They remain foundational for dynamical explanation.
Computer Science and Formal Methods	Supply computation, transition systems, process semantics, protocols, verification, concurrency, and access control. They provide the formal backbone.
Information Science and Visualization	Supply representation, information organization, retrieval, external cognition, and access. The synthesis adds typed behavioral and opportunity consequences.
Learning Sciences and Expertise Research	Supply transfer, adaptation, skill, practice, and distributed learning mechanisms.
Activity Theory and Distributed Cognition	Supply mediated, historical, organizational, and socially distributed units of analysis.
Economics and Decision Theory	Supply choice sets, preferences, utility, risk, games, and policy analysis, but generally abstract from projection and interpretation unless extended.

15. Explanatory Gain

The synthesis provides genuine explanatory gain in four forms.

- Disambiguation: it prevents ontology, semantics, representation, interpretation, opportunity, behavior, and evidence from being collapsed.
- Cross-level linkage: it connects formal system properties with projection, observer access, belief, action, trace, learning, and institutional constraint.
- Comparability: it supplies contracts and typed profiles for comparing implementations without assuming informational or semantic equivalence entails behavioral equivalence.
- Research coordination: it converts scattered constructs into cumulative artifacts that can be tested, reused, and revised.

These are substantive contributions. They justify a named framework and research program. They do not yet prove theoretical autonomy because the gains arise mainly from disciplined integration and boundary management rather than from a uniquely interaction-specific mechanism.

16. Final Falsification of Disciplinary Necessity

Criterion	Assessment
Distinct object	Pass. Computational interaction is a coherent primary explanandum with identifiable episodes, boundaries, and evidence.
Coherent ontology	Pass provisionally. The ontology is internally consistent and cumulative, though many entities are inherited.
Unique mechanisms	Fail. No mechanism has been shown to belong uniquely to computational interaction or to resist explanation by integrated adjacent sciences.
Autonomous methods	Fail. Experimental, observational, formal, psychometric, causal, and predictive methods are inherited and adapted.
Formal independence	Fail. The recommended foundation is explicitly a layered composition of established formalisms rather than a new calculus.
Predictive superiority	Not established. Scoped prediction is possible, but incremental gain over strong integrated baselines remains an empirical hypothesis.
Explanatory necessity	Not established. The synthesis improves coordination and comparison, but current evidence does not show that existing disciplines cannot jointly explain the phenomena.
Cumulative empirical community	Emergent rather than demonstrated. The framework defines a program, but replication, benchmark tasks, shared datasets, and sustained institutional practice remain future work.

17. Research Agenda

23. Publish stable machine-readable schemas for the Interaction, Opportunity, Behavioral Evidence, Measurement, Similarity, Transfer, Strategy, Prediction, and Comparison contracts.
24. Create benchmark interaction episodes spanning GUI, command, API, conversational, multimodal, accessible, collaborative, and safety-critical systems.
25. Develop shared annotation datasets preserving raw records, observation models, attributed events, uncertainty, and rival interpretations.

26. Test reliability of episode boundaries, opportunity coding, event attribution, and strategy inference across laboratories.
27. Evaluate whether typed opportunity and projection variables improve explanation and prediction beyond established HCI, cognitive, control, and process models.
28. Develop causal experiments separating semantic preservation, projection, signaling, capability, permission, and belief.
29. Test transfer and adaptation across representation, modality, device, task, and institutional context using source-target similarity contracts.
30. Formalize compositional mappings among transition systems, event structures, process algebra, probabilistic models, and measurement traces.
31. Establish non-compensatory criteria for accessibility, safety, authorization, and irreversible harm in comparative evaluation.
32. Conduct preregistered multi-site replications before making discipline-level claims.

18. Open Problems

- Boundary ontology: when do analyst-chosen episode boundaries correspond to stable causal or functional organization?
- Reciprocity threshold: what minimum recurrent closure distinguishes interaction from extended one-way causation?
- Emergent action vocabularies: how should open-ended interaction be modeled when available actions are constructed during activity?
- Collective agency: how should participant identity and strategy be represented in distributed human-machine organizations?
- Semantic grounding: under what conditions can semantic state be specified independently of interpretive practice?
- Opportunity measurement: can believed, signaled, and recoverable action relations be estimated reliably across populations and modalities?
- Mechanism identification: can traces distinguish competing cognitive, ecological, institutional, and computational mechanisms?
- Nonstationarity: how should models remain valid when participants, systems, policies, and conventions co-adapt?
- Ethics and power: how should coercion, manipulation, exclusion, and institutional authority enter the ontology without being reduced to preference or performance?
- Disciplinary threshold: what empirical achievements would convert an interdisciplinary synthesis into a scientifically autonomous field?

19. Canonical Vocabulary

Term	Definition
Computational interaction	Interaction in which at least one participant, coupling mechanism, state transformation, or mediating environment is computationally realized.
Computational Interaction Science	The interdisciplinary scientific program that explains computational interaction episodes, their representations, opportunity structures, behaviors, adaptations, measurements, comparisons, and predictions.
Projection	A declared mapping from target structure to an observer-accessible representation, including preserved, transformed, omitted, and introduced properties.
Opportunity	An umbrella analytical profile coordinating typed action-availability relations; not a primitive or synonym for affordance.
Interaction preservation	Preservation of declared coupling, protocol, boundaries, and reciprocal constraint relations under transformation.
Behavioral preservation	Preservation of a declared behavioral relation or distribution under a specified observation model and tolerance.
Measurement equivalence	Evidence that an operationalization has sufficiently invariant meaning and error properties across compared contexts.
Comparability	Satisfaction of the target, boundary, correspondence, observation, measurement, population, and uncertainty conditions required for a defensible comparison.
Explanatory gain	Improvement in mechanism discrimination, unification, intervention, or anomaly resolution relative to a strong baseline.
Predictive gain	Out-of-sample improvement in accuracy, calibration, transport, or decision utility relative to a declared baseline.

20. Conclusion

The research program has crossed the threshold from a collection of intuitions to a coherent scientific framework. It possesses a legitimate explanatory object, cumulative ontology, first principles, formal representation, measurement architecture, falsifiable claims, comparative method, and empirical agenda. It can support rigorous science.

The evidence does not, however, establish the necessity of an independent discipline. The framework's strongest achievements are synthesis, disambiguation, cross-level linkage, and methodological coordination. Its mechanisms, mathematical tools, measurement theories, and empirical methods remain substantially inherited from HCI, cognitive science, psychology, ecological psychology, cybernetics, control, computer science, information science, learning science, and social theory. Predictive and explanatory gains over strong integrated baselines have been proposed but not yet demonstrated through a cumulative empirical record.

Final determination 2: Computational Interaction Science is best understood as an interdisciplinary synthesis.

This conclusion is not a rejection of the framework. It identifies its scientifically defensible status. The appropriate next step is not disciplinary declaration but empirical competition: the synthesis should be tested against the best combined explanations available from adjacent sciences. Independent disciplinary recognition would become justified only if that program demonstrates durable mechanisms, methods, predictions, or explanatory successes that integration alone cannot supply.

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